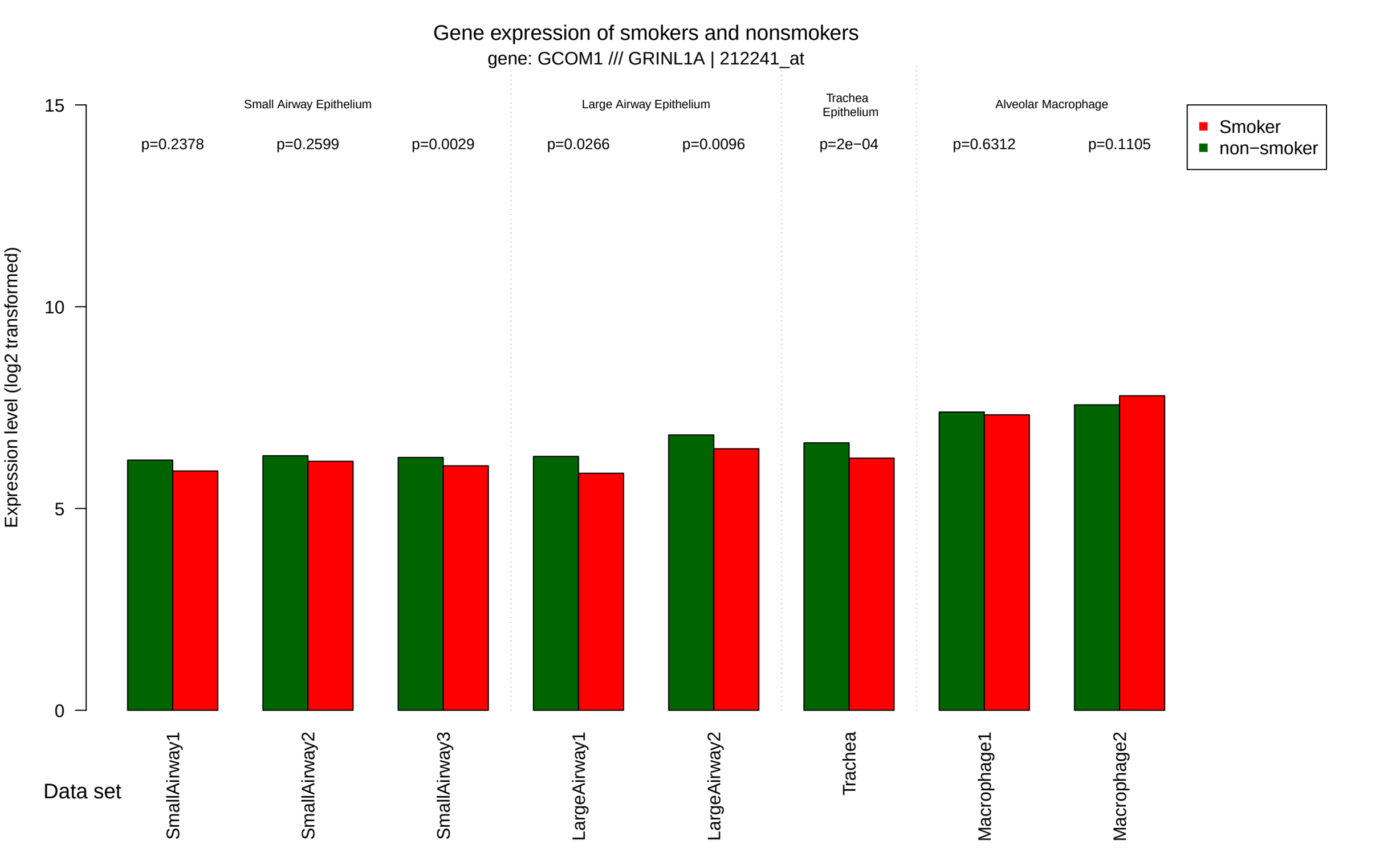
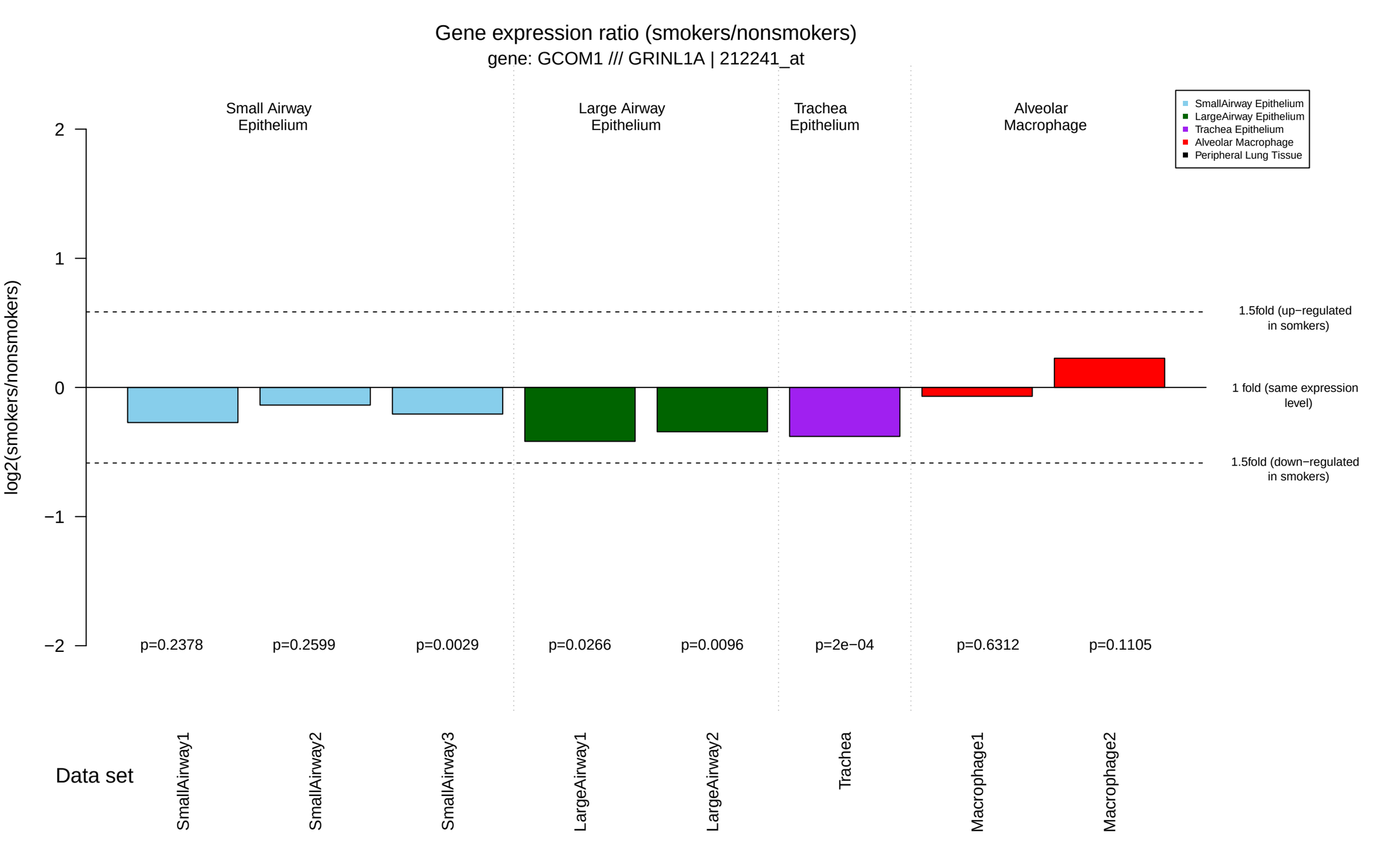
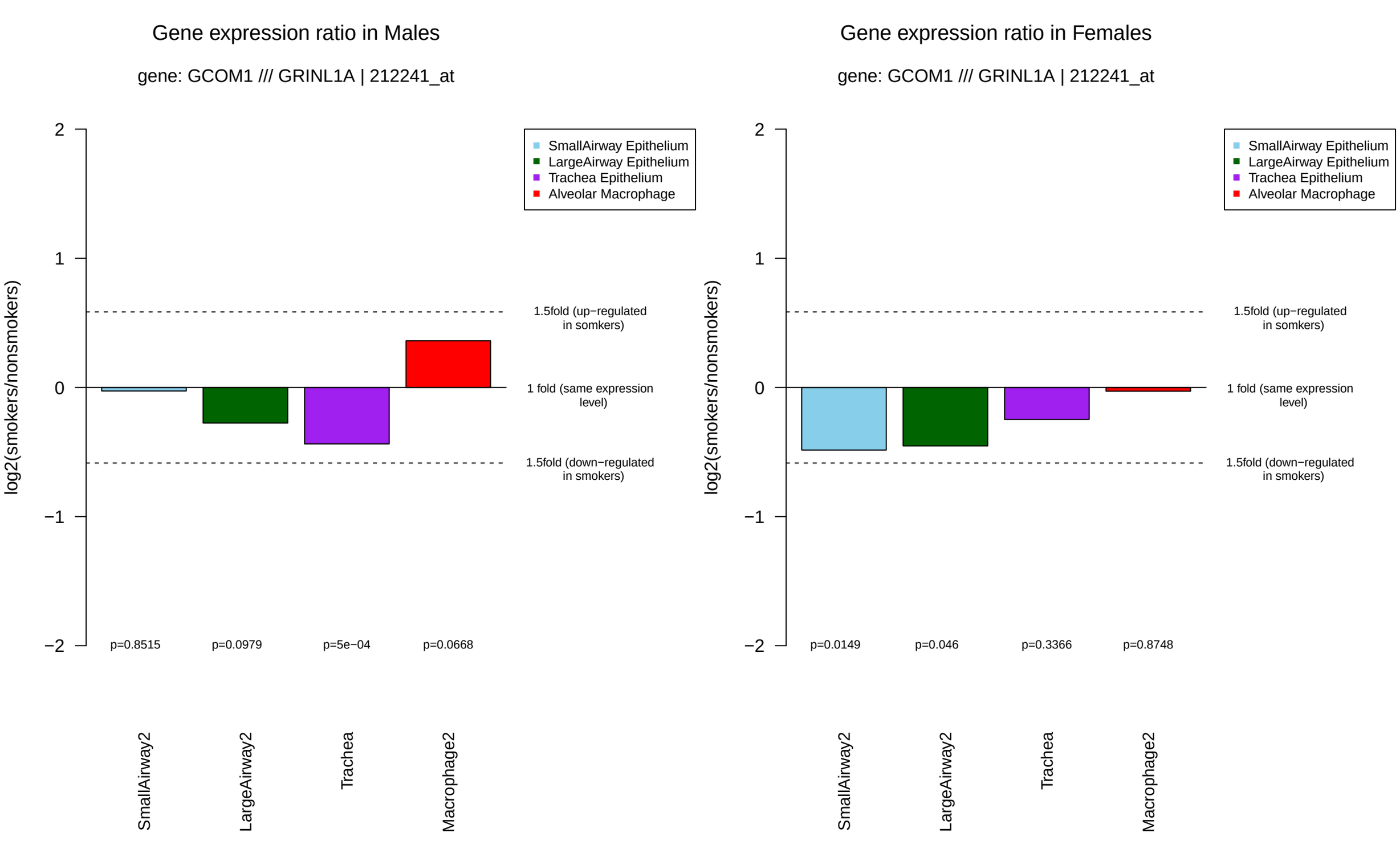
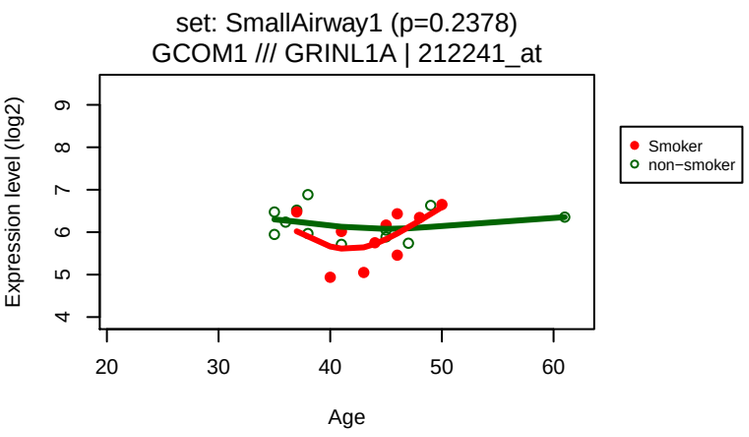
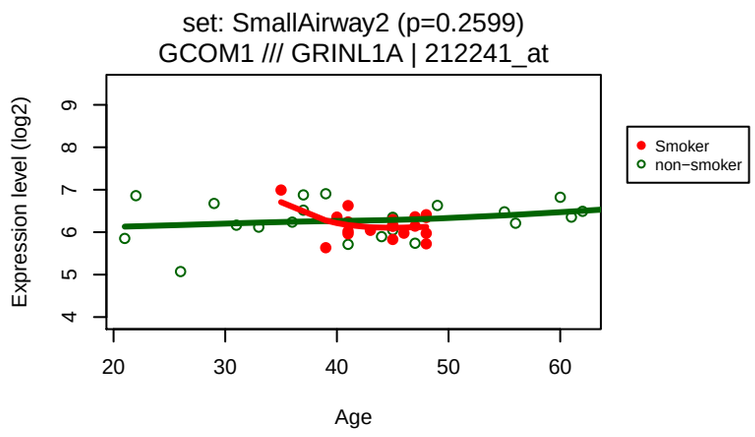
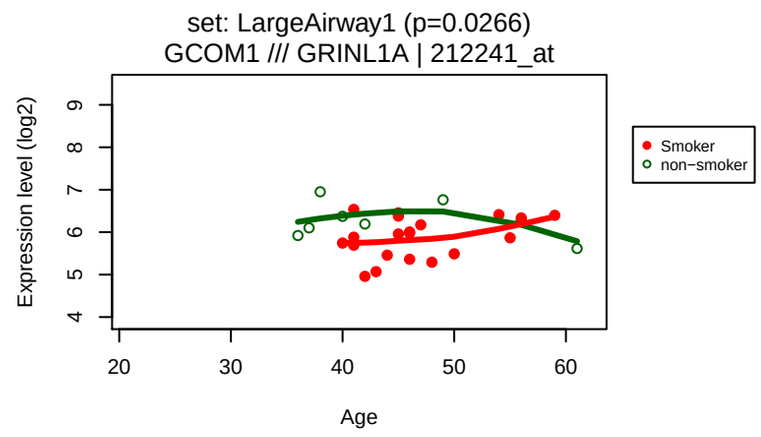

Gene expression of smokers and nonsmokers
gene: GCOM1 /// GRINL1A | 212241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2378
p=0.2599
p=0.0029
p=0.0266
p=0.0096
p=2e−04
p=0.6312
p=0.1105
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

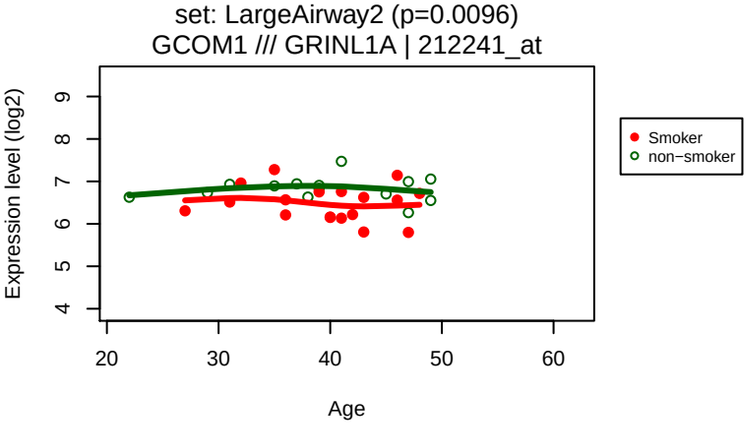

Gene expression ratio (smokers/nonsmokers)
gene: GCOM1 /// GRINL1A | 212241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2378
p=0.2599
p=0.0029
p=0.0266
p=0.0096
p=2e−04
p=0.6312
p=0.1105
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

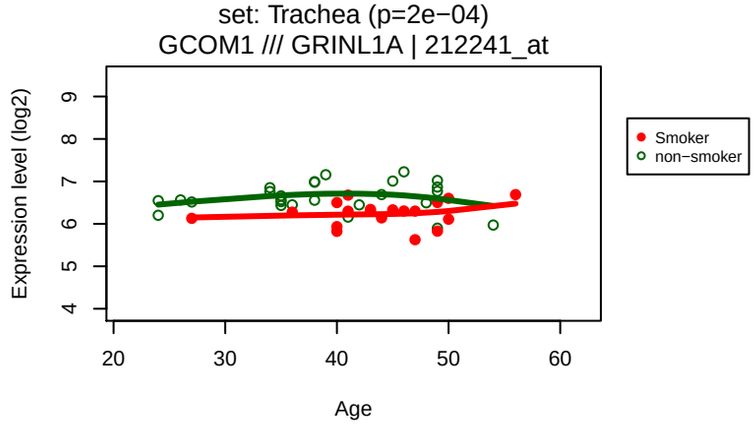

Gene expression ratio in Males
gene: GCOM1 /// GRINL1A | 212241_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8515
p=0.0979
p=5e−04
p=0.0668
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCOM1 /// GRINL1A | 212241_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0149
p=0.046
p=0.3366
p=0.8748
SmallAirway2
LargeAirway2
Trachea
Macrophage2

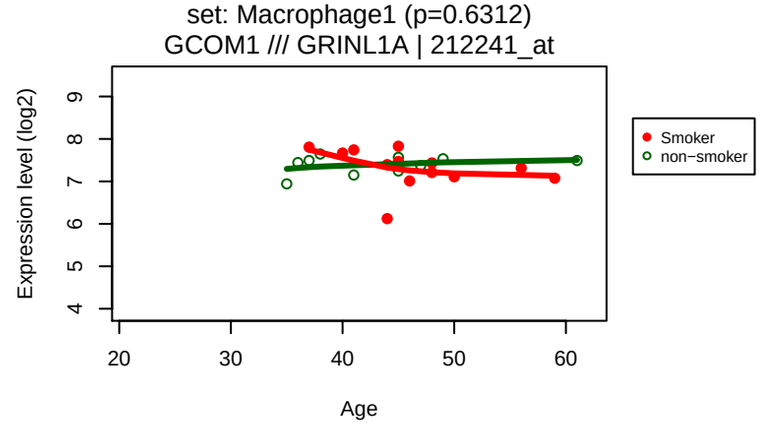

set: SmallAirway1 (p=0.2378)
GCOM1 /// GRINL1A | 212241_at
Expression level (log2)
Age
Smoker
non−smoker

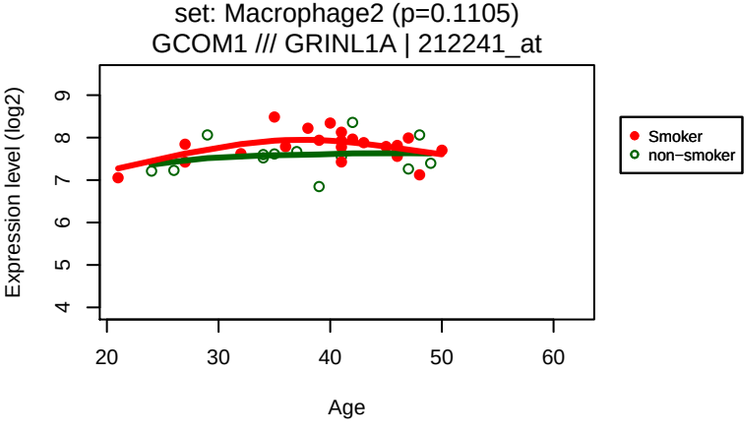

set: SmallAirway2 (p=0.2599)
GCOM1 /// GRINL1A | 212241_at
Expression level (log2)
Age
Smoker
non−smoker

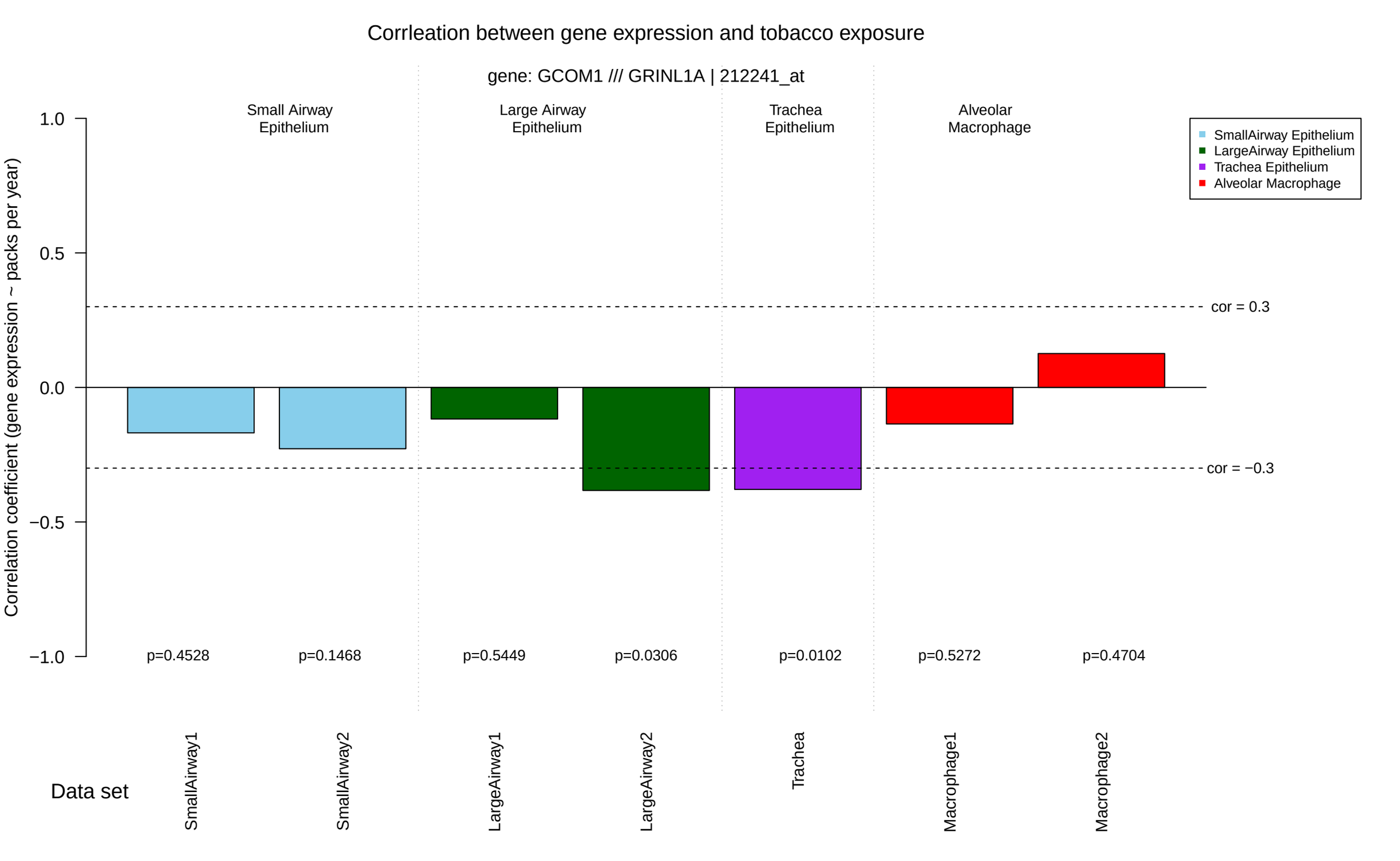

set: LargeAirway1 (p=0.0266)
GCOM1 /// GRINL1A | 212241_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0096)
GCOM1 /// GRINL1A | 212241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
GCOM1 /// GRINL1A | 212241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6312)
GCOM1 /// GRINL1A | 212241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1105)
GCOM1 /// GRINL1A | 212241_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCOM1 /// GRINL1A | 212241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4528
p=0.1468
p=0.5449
p=0.0306
p=0.0102
p=0.5272
p=0.4704
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2